

Kuan-Hung Chen

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Data Science Engineer specializing in Operational Excellence, automated VBA/Python pipelines, and predictive modeling.

Education

University of Warwick

MSc in Statistics (with Distinction)

Focus: Statistical Inference, Data Analysis, Monte Carlo Methods, Machine Learning, Neural Networks (CNN, RNN, LSTM)

Coventry, UK

Sep 2024 – Sep 2025

National Taiwan University

B.A. in Economics & B.A. in Foreign Languages and Literatures

Double Major, Overall GPA: 3.9 / 4.3

Focus: Econometrics, Big Data Marketing, Translation

Taipei, Taiwan

Sep 2019 – Jun 2024

Experience

Market Research Intern

KANTAR Worldpanel (now Worldpanel by Numerator)

Taipei, Taiwan

Jun 2023 – Apr 2024

- Engineered 10+ scalable Python and VBA data pipelines to act as an automation enabler, replacing manual Excel summaries and reporting workflows to drastically reduce operational waste.
- Analyzed complex consumer datasets to extract actionable insights, delivering reports and presentations that provided strategic input to management.
- Managed the end-to-end business solution lifecycle for a primary client account, facilitating cross-functional communication, creating training presentations, and delivering data-driven strategic insights.

Projects

MSc Dissertation (University of Warwick)

Probabilistic Weather Forecasting with Recursive Algorithms and Vine Copulas

Sep 2025

- Engineered an efficient vine copula-based deep learning algorithm utilizing PyTorch, demonstrating advanced quantitative modeling applicable to operational forecasting.
- Implemented a Quasi-Bayesian recursive density estimation algorithm to enable "online" learning and rapid metric monitoring.
- Designed the system to dynamically update predictions upon new data ingestion, streamlining data-driven decision-making in high-variance environments.

Open-Source Developer | obsidian-2-jekyll

Regex-Based Markdown-to-DSL Content Conversion Tool

- Engineered an open-source text-processing pipeline in Python utilizing Test-Driven Development (TDD) to convert complex Obsidian Markdown into Jekyll-compatible outputs for seamless static site rendering.
- Wrote detailed functional specifications and documentation, translating complex regex parsing logic into clear user procedures for open-source adoption.
- Optimized content distribution architecture from raw data ingestion to deployment, maintaining rigorous test coverage and strict version control to guarantee high system reliability.

Machine Learning Project | NLP Product Categorizer

NLP Product Categorizer for zh-TW E-Commerce Context

- Engineered an NLP categorization framework utilizing TF-IDF vectorization and SVC/MNB classifiers, specifically optimized for the complex naming conventions of the zh-TW e-commerce market (mixed English, numeric, and Chinese characters).
- Developed and containerized a production-ready backend API using FastAPI and Docker, enabling seamless deployment and real-time model inference.
- Achieved 70% accuracy and F1 scores against highly noisy, error-prone real-world datasets, establishing a robust performance baseline and demonstrating strong practical viability.

Skills

Programming: SQL, Python (Pandas, NumPy), C++, R, VBA, JavaScript, Ruby, Git

Data Analysis: Stata, Tableau, Advanced Excel, PowerPoint, Time Series Forecasting, ETL pipeline

Machine Learning and Deep Learning: Scikit-learn, CNN, RNN, LSTM, NLP, PyTorch, TensorFlow

DevOps: Bash, Docker, Linux

Languages: Mandarin (Native), English (Professional; IELTS Academic 8.0), German (Intermediate)